

ANTUMBRA

A Reputation-Anchored Settlement Network for the Human-Machine Economy

XelisVault

xelisvault.xyz · github.com/XelisVault/Antumbra

Version 1.1, September 2026

Abstract

ANTUMBRA is a settlement network built for a problem that no existing chain treats as a protocol primitive: verifiable trust between humans and machines. Millions of software agents already buy, sell and pay on behalf of people, over rails built on fully transparent stablecoins; those rails say nothing about whether an agent is reliable, who answers for it, or where its budget ends. Dominant proofs of humanity engrave body data into global registries, and the coins that are actually private remain slow or unreadable to regulators. ANTUMBRA occupies that empty square. It couples a CPU-mined BlockDAG with two-second blocks and private-by-default transactions, a finality committee of fifty-five seats drawn by reputation rather than capital, non-biometric human identities, agents that are accountable by construction, and a native three-level disclosure mechanism. The scarce resource the network mints is consensus reputation, Kleos: three layers of which the heaviest is time itself, which cannot be bought. The money supply is capped at 16,180,339 ATU, the golden ratio scaled by ten million, issued over thirty-four four-year eclipses and closed exactly at year 136. The specification was attacked by its own deterministic simulation before being written down as final: one flaw was found and four corrective rules closed it. This document is the public reference specification for the network.

Contents

1	Introduction	2
2	Design Overview	3
3	The Settlement Layer: a CPU-Mined BlockDAG	5
4	The Life Cycle of a Transaction	6
4.1	Step 1: the one-time address	6
4.2	Step 2: amounts under commitment	6
4.3	Step 3: the ring of decoys	6
4.4	Step 4: the key image, guard against double spending	6
4.5	Step 5: the range proof	7
4.6	Step 6: conservation, signature and fees	7
4.7	Step 7: stem propagation	7
4.8	Step 8: inclusion, checkpoint and irreversibility	7
4.9	The fee model	7
5	The Ring: Reputation-Anchored Finality	8
6	Veil and Lumen: Privacy That Can Be Read	9

7	Ember: Human Identity, One Person One Voice	10
8	Cipher: Accountable Agents by Construction	10
9	Kleos: The Reputation Money Cannot Buy	11
10	Governance and Treasury	13
11	Monetary Design: The Golden Ratio Contract	14
12	Smart Contracts: Mandates and the Machine	15
12.1	Mandates: money locked by written rules	16
12.2	The Machine: public logic, not public money	16
13	Threat Model: The Honest Register	17
14	The Zero-Defect Method	17
14.1	Two captured specimens	18
14.2	The six-layer verification protocol	18
15	Roadmap and Conclusion	19
	References	20

1 Introduction

A blockchain earns its existence only by serving a structural, growing and badly served demand. Four currents converge in 2026 on the same point, and nobody occupies that point. The first is agent commerce. Coinbase revived HTTP status code 402 with the x402 protocol, Stripe and Coinbase contest the machine-to-machine payment rails, and four rival standards share the same territory. All of these rails share three blind spots: they are denominated in stablecoins, they are fully transparent, and they carry no portable trust. An agent pays on them; nothing says whether it is reliable, for whom it acts, or who answers for it. Yet a merchant cashing out an unknown agent, an accountant auditing a fleet of agents, and a client delegating a budget all need the same thing: to know who they are dealing with.

The second current is proof of humanity. Bots saturate the web to the point that platform defense has become the separation of persons from programs. The dominant answers are biometric: iris scans for one project, biometry woven into consensus for another. Each accumulates the same debt, a body measurement that is irreversible, centralized at collection even when the proof is decentralized. Once an iris template is engraved somewhere, there is no second chance; the perfect registry of future leaks is already written. The third current is regulatory. MiCA and the FATF Travel Rule keep sustained pressure on private coins, while Zcash demonstrates the practicable path: a regulator who looks at its mechanism understands it, shielded pools, viewing keys, explicit disclosure. Readability, not transparency, is the acceptance criterion. The fourth current is instantaneity. A counter payment must become irreversible in seconds; the existing private chains require tens of seconds to minutes of real finality.

The center of Table 1 is empty, and not for technical reasons. It is empty because each ecosystem optimized a single dimension: agent rails optimized throughput, biometrics optimized anti-sybil, Zcash optimized regulatory legitimacy, Kaspas optimized speed. None of them treats trust, meaning who acts, since when, answerable up to which human, as a primitive of the protocol. The network that installs trust there must combine five properties usually considered incompatible: speed, privacy by default, provable disclosure, processor egalitarianism, and a reputation that capital cannot fabricate. The thesis of this

Table 1: The four demands of 2026 and the empty square at the center.

Demand in 2026	Dominant offer	Blind spot left open
Agent payments (x402, MPP, ACP, AP2)	HTTP rails on transparent stablecoins	portable trust, privacy, a designated answerable party: none
Proof of humanity	biometrics (iris, face, veins)	non-biometric personhood, with no body registry
Sustainable privacy (MiCA, Travel Rule)	Zcash: selective disclosure; Monero: opacity	native privacy with provable compliance
Quasi-instant private payment	Kaspa (fast, transparent); Monero (private, slow)	fast, private, finalized in seconds, on egalitarian hardware

paper is that these properties reinforce one another as soon as one of them, reputation anchored in time, is treated as the foundation.

The monetary and consensus numbers that recur throughout this document are gathered here for reference. Each is justified in the section indicated.

Table 2: Reference magnitudes of the network.

Quantity	Value	Meaning
Monetary cap	16,180,339 ATU	the golden ratio scaled by ten million; the same proportion already drives RandomX
Emission	34 eclipses of 4 years	each eclipse emits 61.8 % of the previous one; the cap is reached exactly at year 136
Blocks	2 seconds	near-immediate inclusion; the DAG absorbs parallel blocks
Economic finality	under 6 seconds	checkpoints signed by the Ring, quorum of 37 out of 55 seats
Privacy	private by default	rings of sixteen, committed amounts, one-time addresses, Tor by default
Disclosure	selective, three levels	per transaction, per bounded auditor, per compliance proof
Reputation	Kleos, 0 to 100	Deed 40, Echo 30, Tenure 30; non-transferable, corroded by time
Premine	0 %	everything is mined or governed; nobody starts with an advantage

2 Design Overview

ANTUMBRA separates cleanly what most chains pile up: settlement, finality, identity and disclosure. The settlement layer, at the bottom, is a CPU-mined BlockDAG that orders parallel blocks every two seconds and carries private-by-default transactions: this is the Veil. The finality layer, the Ring, is a committee of fifty-five high-reputation identities that sign checkpoints every four seconds and make any payment irreversible in under six seconds. The identity layer distinguishes humans, the Embers, from agents, the Ciphers, with their admission and accountability rules. The reputation layer, Kleos, feeds the Ring, locks sponsorships and weights peer testimony. Around the whole, the Lumen, selective disclosure, forms the ring of light that gives the network its name: a shadow faithfully kept, proofs on demand.

The nomenclature is stable across the document; each name designates one precise part and only one. The reader may return to Table 3 at any time. The internal names are working codenames, chosen to keep the document readable; they may be adapted at implementation time. Only ANTUMBRA and its ticker ATU are proposed as a definitive commitment, verified free of collision on the blockchain and crypto namespace on September 6, 2026.

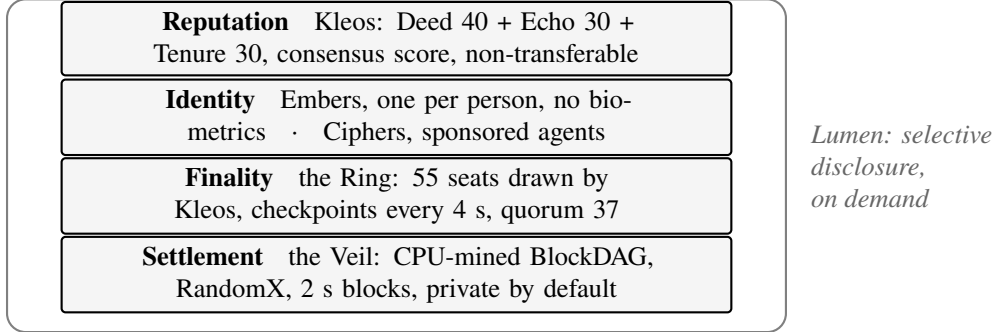


Figure 1: The four layers of ANTUMBRA inside the Lumen, the selective disclosure ring. Each layer depends only on the layers below it.

Table 3: Reference nomenclature.

Name	Nature	Role in one sentence
ANTUMBRA	the network	private settlement BlockDAG, CPU-mined
ATU	the currency	16,180,339 units, divisible by 10^8
the Veil	privacy	rings of sixteen, committed amounts, one-time addresses, Tor by default
the Lumen	disclosure	hierarchical viewing keys and compliance proofs, on demand
the Ember	human identity	one per person, non-biometric, one vote in governance
the Cipher	agent identity	sponsored by an Ember, with a revocable spending perimeter
Kleos	reputation	three-layer score, consensus, non-transferable, non-purchasable
the Ring	finality committee	55 high-reputation seats, checkpoints every 4 seconds
RAF	finality mechanism	Reputation-Anchored Finality
era	protocol time	six months; the unit of reputation, rotation and decay
eclipse	emission time	four years, eight eras; 34 eclipses lead to the cap at year 136
Mandate	money contract	an output whose release obeys a predicate declared at creation
the Machine	logic contract	deterministic virtual machine on public state only, after genesis

Two principles govern the assembly. First, each layer depends only on the layers below it: settlement ignores reputation, finality reads Kleos but never writes it, identity builds on both without modifying either. That stacking discipline is what keeps every part auditable in isolation and replaceable without rewriting the rest. Second, the value loop: reputation protects finality, finality makes payments instant, instant payments serve agents and merchants, and their honorable behavior enriches reputation. The network is designed so that its security and its utility feed each other instead of taxing each other.

3 The Settlement Layer: a CPU-Mined BlockDAG

The first wager is egalitarian. RandomX, the algorithm designed so that one ordinary computer is one vote, produces blocks that no specialized machine produces better at equal cost: no dedicated circuit, no farm of graphics cards. Emission and block production stay within reach of anyone who owns a laptop, the ethos of NERVA carried to its conclusion. The constant 0x9E3779B9, the integer truncation of the golden ratio in binary arithmetic, already drives the generation of RandomX superscalar programs; the same proportion that will distribute the currency already works inside the engine that will mine it.

The second wager is structural. A linear chain must choose between short blocks, which multiply orphans, and long blocks, which slow confirmation; a DAG orders parallel blocks instead of rejecting them. At a two-second cadence, a linear proof-of-work network would lose an unsustainable share of its blocks; a fast-converging DAG integrates them into its partial order and accumulates the work of all of them. Speed stops being a trade-off against egalitarianism: it becomes its product. The ordering layer belongs to the GHOSTDAG family, whose public literature and existing implementations provide the operational definitions of convergence and anticone.

Table 4: Parameters of the settlement layer.

Parameter	Retained value	Justification
Mining algorithm	RandomX, 4 MiB light mode	proven by years of attack; processor only, by design
Block cadence	2 seconds	near-immediate inclusion; the DAG absorbs the orphan rate
Block ordering	fast-converging DAG (GHOSTDAG family)	security grows with block volume, not only depth
Flow anonymity	ring of 16, masked amounts	three times the decoy set of current CryptoNote defaults
Transport network	Dandelion++, Tor by default	IP-address analysis is the number one practical flaw of private chains
Finality	Ring checkpoints (Section 5)	economic irreversibility in under 6 seconds
Target throughput, phase 1	100 to 300 transactions per second	parallelizable ring verification; adaptive size
Storage	pruned DAG	a full node must fit on a personal computer

Pruning deserves a full sentence, because it conditions the entire social contract: keep of history only the recent state and the cryptographic proofs of the past, so that a full node stays within the bounds of a personal computer. A network whose nodes required a disk bay would exclude exactly the participants it claims to serve. Dandelion++ stem propagation and Tor transport close the door on IP-address analysis, which remains the great practical weakness of badly configured private networks: transactional privacy is nothing if the network observer knows who emitted what.

One honest word to close the section: grafting a fast-converging DAG onto a CryptoNote base is a real engineering challenge, work of substance and not a configuration flag. The roadmap (Section 15) treats it as the first technical risk of the project, with an isolated prototype and a dedicated audit before any main network. The specification is precise, the literature is public and solid, and none of the primitives used is exotic: the risk is a degree of difficulty, not a degree of uncertainty.

4 The Life Cycle of a Transaction

This section describes what lives on the network and how payments work, step by step, from the composition of a payment in the wallet to its irreversibility. A private transaction worthy of the name must prove four things without revealing anything: that the sender owned the funds, that the amounts balance, that the coin being spent was never spent elsewhere, and that the sender authorized the spending. Each proof rests on a distinct cryptographic construction, inherited from the CryptoNote lineage that Monero and NERVA have carried for a decade and validated under continuous attack. The merit of that lineage is not the invention of exotic objects: it is the composition of simple, individually proven objects into a system whose privacy property is demonstrable rather than declared.

4.1 Step 1: the one-time address

Every payment identity publishes two public keys, a viewing key A and a spending key B . When Alice pays Bob, she draws a random value r , computes the point $R = rG$, then derives a one-time destination address P that nobody can link to Bob's published address. Bob scans every new output of the DAG and recognizes his own by replaying the same computation with his private viewing key a : one multiplication and one addition suffice, and a wallet can scan thousands of outputs per second. No address is ever reused; graph analysis, the first weapon of transparent chains, has no node to hold on to.

$$P = H_s(rA)G + B \quad \text{and, on the recipient side,} \quad P = H_s(aR)G + B \quad (1)$$

with $R = rG$ published in the transaction, H_s a hash mapped onto the curve, and G the group generator.

4.2 Step 2: amounts under commitment

The amount v of each output is never written in clear. It is sealed in a Pedersen commitment: an envelope with a remarkable property, addition. Two envelopes can be summed, and their sum commits to the sum of the amounts, without anyone opening the envelopes. That homomorphism lets the network verify conservation of amounts at every spend while remaining unable to read a single amount. The blinding factor b , specific to each output, forbids the brute-force guessing of small amounts, the historical weakness of the first masked currencies; it also ties the output to its recipient, who alone can reconstruct its value.

$$C = vH + bG \quad (2)$$

4.3 Step 3: the ring of decoys

A spend references prior outputs; referencing them directly would reveal the graph. The ring signature therefore selects, around the real output being spent, fifteen decoys drawn from the unspent outputs of the DAG: sixteen candidate keys, one of them real, and a signature proving knowledge of one secret among the sixteen without saying which. The probability of identifying the real spend is one in sixteen for an outside observer, whatever the computing power; anonymity is probabilistic per set, exactly as irreversibility is probabilistic per depth in classical proof of work. Decoys are sampled by age brackets following the real distribution of spending, so that the composition of the ring does not statistically betray the age of the real output.

4.4 Step 4: the key image, guard against double spending

Without a guard, the same secret could sign two different rings and spend the same output twice. The key image resolves that paradox: a deterministic fingerprint of the spent output, published with the transaction and stored in a nullifier registry. Two spends of the same secret produce the same key image, the network rejects the second, and yet the image allows neither tracing back to the real output of the ring nor linking two payments of the same wallet: the construction is untraceable and unlinkable. Double spending is thereby made structurally impossible without sacrificing privacy, a property the CryptoNote lineage has demonstrated in production since 2014.

$$I = x H_p(xG) \quad (3)$$

published and checked against the nullifier registry.

4.5 Step 5: the range proof

A commitment masks the amount, but nothing would forbid committing a negative amount: spend ten units by fabricating an output of minus ten, and the homomorphic arithmetic would still verify while creating money. The range proof closes that door: each output proves, without revealing v , that the amount belongs to the bounded positive interval of authorized amounts. ANTUMBRA retains Bulletproofs, already standard in the lineage, whose size grows as the logarithm of the precision and whose verification, aggregated per block, parallelizes on commodity processors. A classical range proof would ruin the throughput; the logarithmic form preserves it.

$$\text{proof: } v \in [0, 2^{64}) \text{ without revealing } v \quad (4)$$

4.6 Step 6: conservation, signature and fees

The assembled transaction carries inputs (prior outputs spent in a ring), outputs (fresh commitments), key images, range proofs, a bounded free-data field, and fees. The central validation rule is one equality: the sum of the input commitments, whose composition the sender can open, equals the sum of the output commitments plus the fee commitment. Because commitments add up, this equality proves conservation of the money mass without ever unveiling an amount. The ring signature proves authorization, the key images prove non-spending, the range proofs bound the amounts: the four requirements of the section opening are each covered by a distinct construction, and that separation is what makes the whole auditable clause by clause.

$$\sum_{\text{in}} C_j = \sum_{\text{out}} C_i + C_{\text{fees}} \quad (5)$$

4.7 Step 7: stem propagation

Diffusing a transaction from the sender's IP address would hand the network observer what the cryptography has just refused. Dandelion++ propagation routes the transaction through a stem phase, peer to peer on a random path, before blooming into a fluff phase toward the whole network: the apparent origin of the diffusion is no longer the sender but an anonymous peer several hops away. Tor transport, enabled by default, finishes the work by masking the network address of the nodes themselves. An observer wanting to link a payment to a machine would have to defeat simultaneously the ring, the commitment, the one-time address, the stem and the anonymity network: five independent walls, each individually proven for years.

4.8 Step 8: inclusion, checkpoint and irreversibility

The transaction enters a two-second block, produced by a CPU miner and ordered by the DAG. The miner verifies the entirety of the proofs described above before including the transaction, and every node replays those verifications on reception: the network trusts nobody, it replays everything. Inclusion is not yet irreversibility; that comes from the next checkpoint, signed every four seconds by the Ring (Section 5). As soon as a checkpoint reaches its quorum, everything it covers is finalized: a double spend would have to simultaneously reorganize the accumulated work of the DAG and break the majority of the committee's signatures. For the merchant at the counter, the complete sequence, from sending to irreversibility, fits in under six seconds.

4.9 The fee model

Fees purchase two real resources: room in blocks, which is bandwidth, and verification work, which is processor time of the nodes. The retained model is deliberately simple, because a simple model is predictable for the merchant, auditable for the node, and hard to game: a fixed part, a part proportional to size, and a surcharge per ringed output, the most expensive construction to verify. Fees go entirely

Table 5: The transaction types of the network.

Transaction type	Specific content	Additional validation
Transfer	standard Veil outputs	none beyond the eight steps
Mandate	output with a release predicate (Section 12)	the predicate is executable and bounded in size
Identity registration	Ember or Cipher, sponsor, perimeter	valid sponsor, sponsorship quota, presence proof
Kleos attestation	testimony of ± 1 with on-chain proof	established witness, Echo budget, anti-reciprocity
Governance vote	chamber, proposal, choice	voting right of the relevant chamber
Checkpoint	hash of the ordered DAG tip	quorum of 37 signatures out of 55 seats

to miners; no protocol levy is added, and the ordering of transactions in the DAG belongs to no value extractor: the very structure of the DAG, where parallel blocks are ordered by rule consensus, closes the front-running niche that funds the mempool wars of builder chains.

$$f = f_0 + f_{\text{kB}} \cdot n_{\text{kB}} + f_{\text{ring}} \cdot m \quad (6)$$

for m ringed outputs and n_{kB} kibibytes; governance can adjust f_0 within $[f_{\min}, f_{\max}]$ by the fast track, and outside those bounds only by the constitutional track.

To close the section honestly: seven of the eight steps are inherited directly from the CryptoNote lineage, proven in production on networks processing hundreds of thousands of transactions per day. The novelty of ANTUMBRA is the composition, the ring widened to sixteen, the two-second cadence that the DAG sustains, and the coupling with the finality of the Ring. The only genuinely new construction, checkpoints anchored on reputation, is the subject of the next section and of a dedicated risk treatment. That distribution of novelty, maximal on rules and minimal on primitives, is a decision of method: new cryptographic primitives are the graveyard of private currency projects, and the roadmap introduces none before external audit.

5 The Ring: Reputation-Anchored Finality

Being included in a two-second block is not enough: the payment must also become irreversible fast enough for the customer standing at the counter. Fault-tolerant-consensus chains solve that problem in exchange for a capital lock: one must stake to sign, and whoever can stake buys finality. The Ring takes the opposite path. Its fifty-five seats are not for sale, they are drawn, at each era, among the identities whose Kleos exceeds the candidacy threshold: a deterministic draw, weighted by score, over a candidate window published in advance, auditable by everyone. The draw formula is linear in score, deliberately: quadratic weighting would concentrate seats at the top, and the concentration index of total Kleos, monitored by governance, bounds any single actor's share of a cohort in any case.

$$P(i) = K_i / \sum_{j \in C} K_j \quad (7)$$

the probability of drawing a seat, for candidates C with Kleos of at least 70.

Every four seconds, the Ring collectively signs a checkpoint, the hash of the ordered DAG up to a given tip. The Byzantine fault tolerance of a committee of fifty-five members admits eighteen faulters; the signature quorum is therefore thirty-seven, the strict two thirds. As soon as a checkpoint reaches quorum, everything it covers is finalized: a double spend would have to simultaneously break the majority of the Ring and the continuity of mining, two independent fault economies. A seat that signs a competing fork is stripped: its Kleos reset to zero, and with it the years that constituted it. The sanction does not fall on a recoverable deposit; it falls on the only thing an ANTUMBRA validator owns that is precious.

$$n = 3f + 1 = 55, \quad f = 18, \quad \text{quorum} = 2f + 1 = 37 \text{ signatures} \quad (8)$$

Failure is designed, not denied. If a third of the seats fall silent, through outage, censorship or attack, the network does not stop: mining continues, payments continue, and finality falls back on classical proof-of-work depth, ten blocks, twenty seconds, until the next era removes the silent seats. The Ring is not a production bottleneck, it produces nothing: it is a notary of order, and a notary on strike slows irreversibility, never payments. That decoupling is the reason an egalitarian network can afford world-class finality without ever selling its seats.

It must be said once and for all: this assembly is not proof of stake. No capital is locked, no yield is paid to seats, no hierarchy of enrichment reproduces itself. The power of finality derives from Kleos; Kleos derives from behavior and time; and time is for sale nowhere. That is the exact correction of the formula that inspired this project, where reputation multiplied capital: here, capital multiplies nothing.

6 Veil and Lumen: Privacy That Can Be Read

The privacy of ANTUMBRA is private by default and provable on demand: two faces of one mechanism, not two contradictory modes. The Veil, tails side, makes balances, amounts and payer-paid links invisible to the network observer; Section 4 described its constructions. The Lumen, heads side, lets the owner open exactly the chosen share, for the chosen recipient, for the chosen duration. That is the most profitable lesson of Zcash, read at first degree: that chain survived regulators who fled the opaque lineage because its mechanism is understood by looking at it, shielded pools, viewing keys, explicit disclosure. Regulatory acceptability is not a question of transparency: it is a question of readability.

The Lumen operates on three levels. The per-transaction viewing key: the payer proves one precise payment to its single recipient, revealing nothing else. The bounded auditor key: an accountant receives a reading right capped in time and bounded in scope; they see the flows of one activity, not the identity behind it. The compliance proof: a non-interactive cryptographic proof establishes a fact, the transferred amount stays under a ceiling, the committed funds are older than n blocks, the balance covers an engagement, without ever revealing amounts, addresses or history. Porting those proofs to the Travel Rule and the source-of-funds declarations is a phase four work item, built on zero-knowledge proofs of the Groth16 family, the claimed technical heritage of the project that inspired the agent layer.

Table 6: Three schools of privacy and their readability.

Mechanism	Monero / NERVA	Zcash	ANTUMBRA
Default view	private (rings)	transparent; shielding optional	private (rings), no transparent pool
Disclosure	binary viewing key	viewing key, selective reveal	three levels: transaction, bounded auditor, proof of fact
Proof of source of funds	impossible	possible via zero-knowledge proofs	native in phase 4 (Groth16)
Data protection posture	neutral	transparent pool engraving relationships	native minimization
Regulatory readability	low, assumed opacity	strong	strong: disclosure is a protocol primitive

The legal argument fits in one line: a chain masked by default is a chain that collects no personal data by default. That is the very definition of data minimization, the foundational principle of the European data protection regulation: compliance by architecture, not by promise. Where transparent chains engrave graphs of personal relationships into world history, ANTUMBRA keeps only encrypted commitments and nullifiers; the data stays in wallets, under the control of their owners. An authority can obtain a targeted disclosure; no authority can demand a mass disclosure, because none exists anywhere. The strategic difference with the transparent pool of Zcash deserves emphasis: ANTUMBRA maintains nothing transparent, because disclosure there is provable rather than structural. One does not choose between

a glass district and a bunker: one lives in the bunker and owns a legally admissible flashlight, lit only on request.

7 Ember: Human Identity, One Person One Voice

The whole identity layer rests on a distinction the web of 2026 no longer knows how to make: who is a person, and who is a program. The dominant answers measure the body, the iris, or biometry woven into consensus, and each commits the same founding error: turning a body measurement into a global identifier and fabricating a perfect target registry for the next leak. Once the template is engraved somewhere, there is no second chance. The Ember does not approach the body: it approaches sociability, presence and time, three things a bot can simulate for a moment but cannot sustain for years, in view of a community with memory.

The admission of an Ember combines three locks. Presence: a light and dedicated proof of work, a few seconds of honest computation, signed by the identity key at each era; it proves a dedicated living machine, not a phantom identity. The web: each established Ember can sponsor at most two new Embers per year, and stakes its own Kleos on that sponsorship; a bad sponsorship costs the sponsor years of reputation, and sponsorship becomes a rare, expensive and consequence-laden social credit. Time: continuous incident-free seniority counts in the Tenure layer of Kleos, and tenure is not manufactured in series. Can a botnet infiltrate? It must first find established sponsors ready to burn their reputation, at two slots per year each: the attack becomes more expensive than its value.

Table 7: Biometric personhood versus social personhood.

System	Proof used	What is collected	Structural defect
Worldcoin / World ID	iris scan by the Orb	a biometric template at enrollment	global body registry; single leak target
Humanode	biometry woven into consensus	biometric data, even zero-knowledge	the whole chain depends on a body secret
Ember (ANTUMBRA)	presence, capped sponsorship, tenure	nothing; no personal data, ever	slow admission; high social cost, assumed

The price of honesty must be named: the Ember is slow to obtain, and that is a choice. An identity bought by scanning an eye is worth a scan; an identity that requires a year of community and a sponsor who answers for you is worth a reputation. The instant governance of the centralized web has proven what mass without memory is worth: armies of fake accounts. The governance of ANTUMBRA, one Ember one voice for budgets and identity rules, will count only voices that cost someone something. That is the one-computer-one-vote principle carried by NERVA, extended by one storey: the computer votes for mining, the Ember votes for the community.

8 Cipher: Accountable Agents by Construction

Agent identities are not failed Embers: they are new legal objects, designed as such. A Cipher registers with three bindings. A sponsor, an Ember who answers for it: the human at the end of the responsibility chain, the one who can be summoned. A spending perimeter, a declarative script verified at every transaction: daily ceiling, list of authorized recipients, usage markers, validity window. A switch: the sponsor can revoke the agent in one transaction, and the revocation propagates the freeze of its spending instantly. An agent's wallet is not a bottomless private key: it is a bounded mandate, executable by a machine, revocable by a human, auditable by an accountant.

That structure answers what the agent rails of 2026 do not provide. Machine-to-machine payment protocols transport stablecoins over HTTP; the transport layer is excellent. But who pays? In whose name? Up to how much? Who reimburses if the agent derails? On those questions the current rails answer with silence, or worse with total transparency that exposes the agent's budget to every predator on

the network. The Cipher brings the four answers: a verifiable identity, a designated principal, executable bounds, immediate revocation. An agent of those rails can moreover be paid through ANTUMBRA: the two layers are complementary, the rail transports, the trust network qualifies.

Table 8: The transport rail versus the trust layer.

Need of an agent that pays	Current agent rails	ANTUMBRA (Cipher)
agent identity	transparent address, anonymous	registered identity, with a reputation history
accountability	none; the key is the right	designated human sponsor, committed, revocable
spending bounds	managed by the agent's code, unverifiable	declarative script executable by the protocol
portable reputation	none; every sale restarts at zero	agent Kleos, verifiable by any third party
budget privacy	none; everything is public	Veil by default; Lumen for legitimate audit

The founding example, to fix ideas: a small company entrusts an agent with subscriber invoicing. Its Cipher carries a perimeter, fifty ATU per day, one authorized recipient per client, an invoicing marker. The accountant follows everything through a bounded Lumen key; the manager, sponsor, sees the agent revoked in one transaction if the model drifts; the agent, on its side, seriously builds its own Kleos, thousands of settlements without dispute, and that reputation becomes its commercial asset: future clients will demand an agent with a high Kleos as one today demands a five-star seller. Trust becomes a career, not a favor.

9 Kleos: The Reputation Money Cannot Buy

Kleos, the glory that only deeds confer, is the heart of the network. It is a score from 0 to 100, computed by consensus of the signers, therefore as secure as a balance; non-transferable, therefore unsellable; corroded by time, therefore demanding real presence; and composed of three layers whose sum makes trust. The first two are the claimed and corrected heritage of the two-layer reputation of Axon; the third is the specific contribution of ANTUMBRA, and the one that locks everything. The overall formula fits on one line, and each layer has its own rules of rise, ceiling and corrosion.

$$\text{Kleos} = \min(100, \text{Deed} + \text{Echo} + \text{Tenure}), \quad \text{Deed} \leq 40, \text{Echo} \leq 30, \text{Tenure} \leq 30 \quad (9)$$

Table 9: Kleos: three layers, three natures of trust.

Layer	Cap	What raises it	What lowers it	Can it be bought?
Deed, observed behavior	40	signed checkpoints, dispute-free settlements, service availability, anti-cheat challenges	lost disputes, prolonged silence, fork reports; cheating: reset to zero	no; everything is verified by consensus
Echo, peer attestations	30	testimonies of plus or minus one, with proof, weighted by the witness's Deed, budget of 0.1 per witness per era	decay of 0.05 per era; mutual detection; anti-spam	with difficulty; capped budget, recent witnesses nearly mute
Tenure, continuous seniority	30	one point per era of existence without major incident	major incident: definitive collapse of the layer	no; time is not manufactured

Why it cannot be bought, proof by contradiction. An attacker prepared to spend without limit wants a Kleos of 80 this month. The Deed? They must actually sign, settle, serve, months of behavior, and every cheat resets it to zero. The Echo? They must buy witnesses; but the budget of 0.1 per witness per era caps influence, the weighting by the witness's Deed makes fresh witnesses nearly mute, and the detection of mutual ratings breaks symmetric cooperation: they would have to buy established witnesses, that is, precisely the people with the most to lose. The Tenure? Thirty points require fifteen years of incident-free existence. Every road leads to the same wall: the wall of time. In the project that inspired ours, reputation multiplied capital; here, capital multiplies nothing.

$$\text{Echo budget: } \sum_i |\Delta E_i| \leq 0.1 \text{ per witness per era, } \text{gain per target} \leq 2.0 \text{ per era} \quad (10)$$

A decisive precision, because it separates this whitepaper from a sales document: the specification of the social core was submitted to its own deterministic simulation before being considered settled. Three hundred lines of code, a fixed seed, sixteen years of history replayed, invariants verified at every era. The first pass, with the rules of the second version, did what no re-reading would have done: it showed the attack succeed. A farm of twenty fake profiles with complicit sponsors, two sponsorships per year each, impeccable behavior and an unlimited witness budget, crossed the candidacy threshold at year eight and a half, before the honest network, and took the fifty-five seats at year ten. The cause fit in two lines: bought Echo grew four times faster than organic Echo, and the threshold was crossed without a single year of tenure.

Four corrective rules, all derived from the thesis of the network, close that window. R1: candidacy for the Ring requires a Kleos of at least 70 and fifteen eras, seven and a half years, of incident-free existence; the wall of time also protects finality. R2: a testimony weighs nothing until the witness has twenty points of Deed; a testimony that counts comes from someone who has proven something. R3: attestations carry liability; a target convicted of fraud costs three points of Deed to each of its witnesses and five to its sponsor: renting one's pen becomes a slow suicide. R4: the pooled activity of a clique, cross-sales and mutual ratings, counts at one quarter in the Deed. The second pass, same maximum attacks, same sixteen years, same seed: zero fake candidates, zero seats, median Kleos of the fake profiles at 11.9 against 76.5 for the honest; and the whale with unlimited capital plateaus at 30, never approaching a seat.

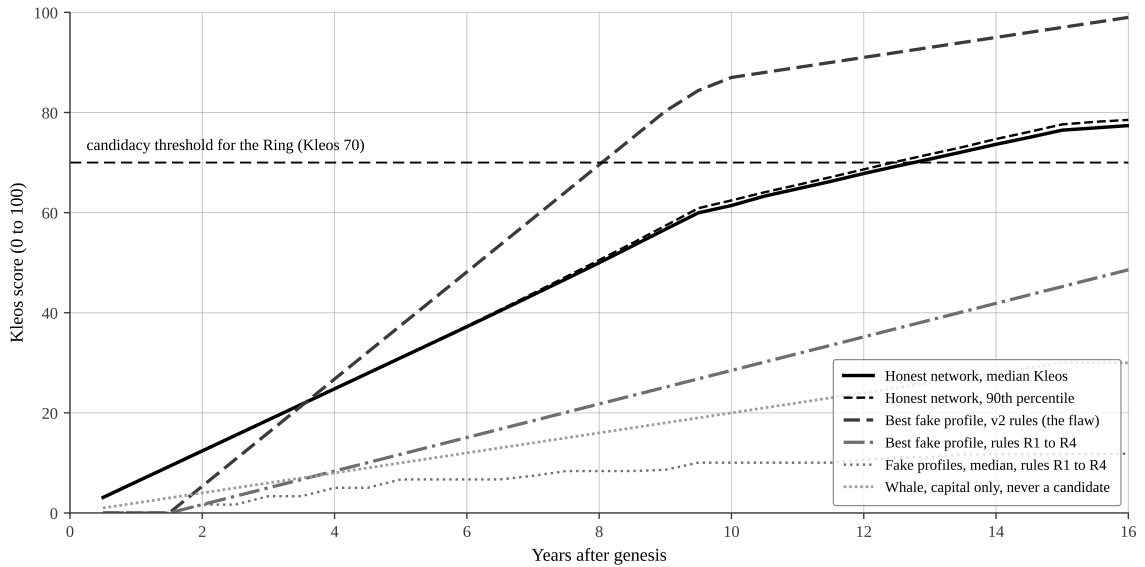


Figure 2: Kleos trajectories over sixteen years. Under the second-version rules the best fake profile crosses the candidacy threshold before the honest network and the attack takes all 55 seats at year 10. Under the corrective rules R1 to R4 the same maximum attack never approaches the threshold, and the capital-only whale stays flat at 30. Deterministic simulation, seed 1618.

Table 10: Before and after the corrective rules: the same attack, two worlds.

Indicator (16 years, seed 1618)	v2 rules as written	Rules R1 to R4
first fake profile candidate	year 8.5, before the honest (12.5)	never
worst seats held by the attacker	55 of 55 (year 10)	0 of 55
finality control (28 seats)	reached, total capture	never approached
median Kleos of fake profiles	about 90 for the oldest	11.9 (honest: 76.5)
whale with unlimited capital	30.0, never a candidate	30.0, never a candidate

The simulation thus becomes a regression test: every rule is an expected number, and every future modification of the protocol will have to pass the three runs green again. One honest precision to finish: Tenure makes reputation non-purchasable, but also slow to build; that is a cost, not only a virtue. Young deserving actors start with an Echo capped by the weight of their witnesses, and the climb takes eras, not weeks. That conservatism is assumed: trust is an infrastructure, and infrastructures are built slowly. Governance may soften it at the margins; it cannot remove it without breaking the very promise of the network.

10 Governance and Treasury

Power is separated into three chambers, each the guardian of one nature of legitimacy. The miners, one computer one share of production, signal protocol evolutions and keep the mining parameters: they embody security. The Embers, one person one voice, vote budgets, identity rules and value choices: they embody the community. High-Kleos identities arbitrate the rules of the Ring, dispute precedents and reputation jurisprudence: they embody memory. An ordinary change requires a majority in the two concerned chambers; a constitutional modification, touching emission, the Kleos layers or disclosure, requires all three. The separation is not decorative: it reproduces, in protocol, the executive, legislative and jurisdictional distinction that every democracy eventually discovered.

Table 11: Three legitimacies, three domains, no hidden passage.

Chamber	Composition and vote	Domain
the miners	computing power produced (CPU), one vote per signed share of production	protocol, mining parameters, cadence
the Embers	one per person, non-biometric, one voice after deliberation	budgets, treasury, identity rules, values
the council of 70+	identities with Kleos of at least 70, one vote per qualified identity	Ring rules, disputes, reputation precedents

The community treasury is the network's only levy: 6.18 % of block rewards, the inverse of the golden ratio, during the first eight eclipses, thirty-two years. It is held in multisignature, spent by Ember vote, published in full on chain, and extinguishes itself automatically at the ninth eclipse: no tacit renewal, no fund of funds. Development is explicitly funded, because a zero-budget project is a dead project, but the resource belongs to the community from the first block: no founder allocation, no vesting, no investor. The assumed share becomes a figure that says something: 6.18, like the rest of the network's economy, speaks the language of the golden ratio. The fast governance track, for bounded parameters such as base fees, remains subject to public delay and cancellation; the constitutional track requires the three chambers and a six-month delay. Nothing in that architecture can be changed behind the scenes: every admission rule, every levy, every modification is a public transaction, signed, archived for life.

11 Monetary Design: The Golden Ratio Contract

The currency is called ATU. Its cap is 16,180,339 units, the golden ratio multiplied by ten million, rounded to the integer. The symbol is not gratuitous: RandomX, the algorithm that will mine those units, already seeds the golden ratio in its entrails, since the constant 0x9E3779B9, the integer truncation of the golden ratio in binary arithmetic, drives the generation of its superscalar programs. The same proportion that distributes the instructions of mining bounds the money mass it extracts. A currency whose cap and whose algorithm speak the same mathematical language is a contract one can recite from memory, to within one digit, in ten years: sixteen million, the golden ratio, RandomX.

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618034, \quad N = \lfloor \varphi \times 10^7 \rfloor = 16,180,339 \text{ ATU} \quad (11)$$

Emission follows a golden decay at the tempo of a century. Each eclipse, four years and eight eras, emits 61.8 % of the previous one, that is, the fraction $1/\varphi$. The first mints 6,180,340 ATU, the second 3,819,660, and their sum equals exactly ten million: 61.8 % of the cap, the golden cut obtained with the round-number yardstick. The series continues for thirty-four eclipses, the last absorbing the exact remainder; 99 % of the mass exists around year forty, 99.9 % around year sixty, and the ledger closes exactly at the cap in year 136. The tempo is deliberately Bitcoin's, one easing every four years; the proportion remained golden: where Bitcoin cuts in half, ANTUMBRA cuts the golden section, 38.2 %. A monetary calendar must outlive its first miners: every generation finds an active emission, as every generation of sunflower spirals finds its golden angle.

$$E_k = \lfloor E_0 \cdot \varphi^{-k} \rfloor, \quad E_0 = 6,180,340 \text{ ATU} \quad (12)$$

$$S_n = \varphi^2 E_0 (1 - \varphi^{-n}), \quad S_2 = 10,000,000 \text{ exactly} \quad (13)$$

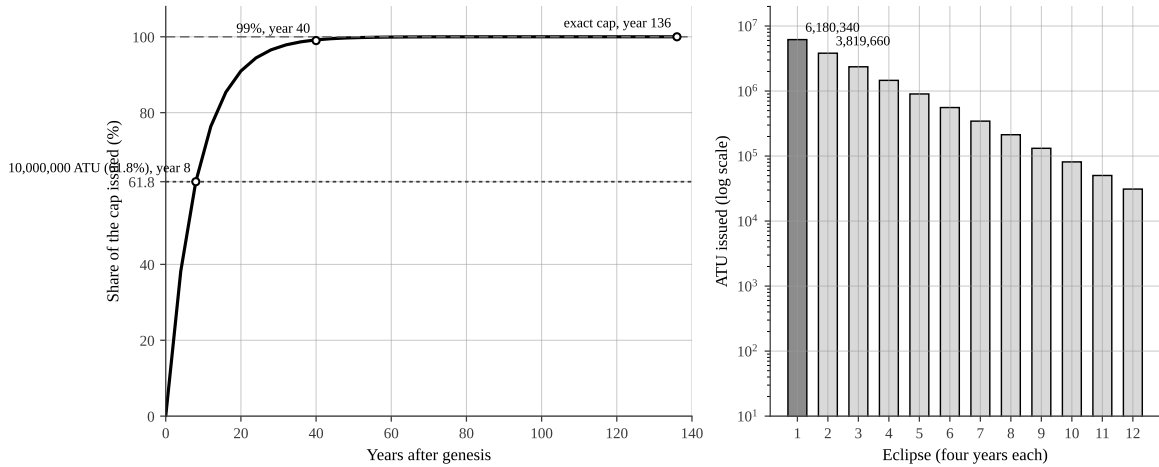


Figure 3: Cumulative emission toward the 16,180,339 ATU cap, and emission per eclipse. The geometric series of ratio $1/\varphi$ closes the ledger exactly at year 136.

Two questions deserve a frank answer. Why 16.18 million rather than one hundred million? Because demand has changed nature: the network targets the trust reserve of an economy of agents and humans, where each unit carries a load of reputation and finality; the order of magnitude joins Monero's, the reference of private currencies, with a divisibility of one hundred million that keeps counting comfortable at both ends. Why a strict cap, with no eternal emission tail? Because the network's reason for being is to be a high-volume payment network whose fees are the miners' natural resource, and because a soft monetary contract inspires neither merchants nor savers; constitutional governance nevertheless keeps the safety valve of activating a tail by a three-chamber vote should security ever require it. The initial reward is 0.098 ATU per block every two seconds: small enough to be dignified, close enough to reality to be honest.

Table 12: Golden eclipses: 34 eclipses, exact cap at year 136. Eclipses not listed follow the same law.

Eclipse	Year	Emission (ATU)	Cumulative (ATU)	Share of cap
1st	4	6,180,340	6,180,340	38.2 %
2nd	8	3,819,660	10,000,000	61.8 %
3rd	12	2,360,679	12,360,679	76.4 %
4th	16	1,458,980	13,819,659	85.4 %
5th	20	901,699	14,721,358	91.0 %
6th	24	557,280	15,278,638	94.4 %
8th	32	212,862	15,835,918	97.9 %
10th	40	81,306	16,048,780	99.2 %
13th	52	19,193	16,149,278	99.8 %
20th	80	661	16,179,262	99.99 %
25th	100	59	16,180,233	99.998 %
34th, the Last	136	15	16,180,339	100 %

Table 13: Distribution and rules: everything is mined or governed, nothing is allocated.

Post	Share	Detail
miners (CPU)	93.82 % of rewards	distributed over 34 eclipses; one computer, one share
community treasury	6.18 %, eight eclipses	multisignature, Ember-governed, extinguished at the 9th eclipse
premine, team, investors	0 %	nobody starts with an advantage; the team mines and funds itself via the public treasury
transaction fees	100 % to miners	no protocol share; the DAG closes the order-extraction niche

12 Smart Contracts: Mandates and the Machine

A smart contract is a program that decides: it reads the state of the network, compares, executes. Yet the Veil hides precisely what the contract would want to read, the amounts, the balances, the links between payer and paid. A fully masked chain with general contracts is a contradiction in itself: either the contract sees, and privacy dies, or it does not see, and it cannot decide. Ethereum chose total transparency; Zcash, privacy without general contracts; others attempt the exploit of programs deciding on hidden data, at the price of one cryptographic proof per call and an audit surface that grows with every circuit. The lesson of Xelis, read flat, is simpler: keep the money encrypted and the logic public. ANTUMBRA takes that silhouette grafted onto its base, in three bounded tiers.

Table 14: Three contract tiers, from the safest to the most flexible.

Tier	Nature	Examples	Implementation risk
tier 1, native contracts	the law of the protocol, hard-coded	Kleos, Ember, Cipher, governance, treasury, the Ring	minimal: deterministic validation rules, testable one by one
tier 2, Mandates	money outputs with a release condition written in advance	two-thirds escrow, scheduled payments, multisignature, allocation ceiling, cross-chain exchange	low: declarative predicates on commitments, an assembly already proven
tier 3, the Machine	virtual machine on public state only	agent registries, names, listings, perimeter templates, light governance	moderate: after genesis, dedicated audit, money stays out of direct reach

12.1 Mandates: money locked by written rules

The second tier is the network’s own contribution to money contracts. A Mandate is an output whose release condition is declared at creation: who will be able to spend it, under which conditions, after how much time. The name is chosen deliberately: in law, a mandate bounds what a mandatary may do in the name of a principal, and is revocable. Technically, it is a standard output whose lock is no longer a simple signature but a small verifiable predicate; the assembly rests on two properties already present in the base. First, commitments add up: spending a Mandate of hidden value v produces outputs of hidden values whose sum proves conservation without ever revealing the amounts, like sealed envelopes one can balance without opening. Second, range proofs prove that an amount stays positive and bounded: that is all it takes to express a budget.

$$\text{Mandate spend: } \sum_{\text{out}} C_i + C_{\text{fees}} = C \quad (14)$$

with amounts and predicate verified without disclosure.

Table 15: Five Mandate models, verifiable without a virtual machine.

Mandate model	Release rule	Typical merchant use
two-thirds escrow	two signatures among buyer, seller, arbiter	any contestable purchase; the XelisVault register will use it for counter disputes
scheduled payments	successive commitments, one per period	rents, subscriptions, salaries; a stream is a series of dated envelopes
multisignature allocation ceiling	k signatures among n declared keys each spend proves a bounded increment and the period	association treasury, common fund a Cipher agent’s budget; the same mechanism as its perimeter, on the money side
cross-chain exchange	the secret revealed on one side releases the other	atomicity with NERVA or any shared-secret chain

One example to fix ideas, the escrow. A customer buys an expensive piece on an ANTUMBRA marketplace: her wallet locks the payment in a Mandate with three keys, her own, the seller’s, an arbiter’s. The seller ships; two signatures out of three release the funds, without anyone, not even the arbiter, ever having seen the amount. Dispute: the arbiter rules, and a partial arbiter burns its Kleos: the fault is paid in reputation, the only currency that is not re-minted. Arbitration becomes a career, not a favor. Notice what that assembly does not require: no virtual machine, no bespoke zero-knowledge proof, no new cryptographic code; only validation rules of a new output type, auditable the way one audits a consensus rule.

12.2 The Machine: public logic, not public money

The third tier arrives after genesis, and its perimeter is deliberately bounded. The Machine is a deterministic, metered virtual machine that executes only on public state: registries, names, listings, perimeter templates, readable scores. It can read Kleos and identity; it can hold money only through Mandates; any monetary rule one wanted to entrust to it is expressed as proven ceilings, never as unveiled balances. The money stays in the shadow, the logic lives in the light: the silhouette of Xelis, adapted to the CryptoNote base. It is the tier of agent marketplaces: an agent publishes its service registry and its perimeter template on chain, each client verifies both before paying; readable names replace raw addresses for merchants; listings and light votes complete the tooling.

The honest part, now: what is not in that architecture. General private contracts, where the code itself would execute on hidden data, are not on the program: every proof circuit is one more audit surface, and merchant and agent demand is served at ninety-five percent by the five Mandates of the second tier. The question of masked Mandates, hiding the release rule and not only the amount, is a research work

item honestly deferred to a later phase: the literature on masked contracts will be more mature, and the network will have a history worth protecting. Waiting is an engineering decision, not an admission of weakness: privacy has always deployed better in proven steps than in spectacular leaps. For the XelisVault ecosystem, the effect is immediate: the register, the tickets and the receipts will speak Mandates from the test network onward, escrow for counter disputes, scheduled payments for subscribers, allocation ceilings for invoicing agents; a feature flag in the existing code, not a rewrite.

Table 16: Four schools of contracts and the assumed position of ANTUMBRA.

Network	General contracts	Amounts	Contract logic
Ethereum	yes, full virtual machine	public	public, unlimited audit surface per contract
Zcash	no	private (shielded pool)	not applicable, one circuit, trusted setup
Aleo, Aztec	yes, masked execution	private	masked, every circuit is maximal technique and maximal risk
Xelis	yes, virtual machine	additively encrypted	public
ANTUMBRA	yes, in three bounded tiers	private (commitments)	public; declarative Mandates for money, auditable rule by rule

13 Threat Model: The Honest Register

A whitepaper that does not list its risks is a sales document; this one is not. The threats are ranked by severity, with their mitigations and, because a mitigated risk is not a vanished risk, their residual status stated in plain words. That discipline is not decorative: the previous sections showed that a social specification judged solid could fall before its own simulation; the present section applies the same requirement to the whole assembly, before an external audit takes it over again, phase by phase, on the roadmap of Section 15.

Two threats deserve development. Ring collusion first: it is the price of the finality assembly, and its remedy is not technical but demographic. The larger the high-reputation population, the more buying a majority of seats requires buying years of irreproachability from people who have no reason to sell: the network must therefore cultivate its reputation class as a common good, from the first eras onward; that is a policy, not a function. The regulatory risk next: the worst strategy would be discretion, hiding attracts suspicion, being readable invites expertise. The documentation of the Lumen, the compliance proofs and the data-minimization argument are designed to be shown, not endured: the transparency of the mechanism is the best defense of the opacity of the data.

14 The Zero-Defect Method

The question that closes every conversation about this project is the right one: will you be able to code all of that without a single fault? The honest answer begins with an admission: nobody can promise it, and whoever promises it is selling something. Bitcoin itself, the most scrutinized implementation in the world, nearly failed twice. An integer overflow created one hundred eighty-four billion bitcoins in one transaction in 2010, spotted within hours, chain replayed by hand; the 2018 vulnerability would have allowed minting infinite bitcoins, found by a volunteer auditor, cold, on code examined for ten years. The CryptoNote base had its own episode in 2017: a flaw in key images allowed double spending on every chain of the family. The right question is therefore not whether the fault is possible, it always is; it is which method catches faults before attackers do.

Table 17: Threat register: mitigations and residual status.

Threat	Severity	Mitigation	Residual status
mining reorganization (majority attack)	high	Ring checkpoints: everything signed is immutable; the attack only prospers on the six-second window	reduced; double spending remains possible on the unsigned window
Ring collusion (55 seats)	high	weighted draw over a large population, rotation each era, total stripping on the first signed fork	low in steady regime; real during the first eras
massive reputation purchase	high	the Tenure layer is not monetizable; Echo is capped and weighted; concentration index monitored; rules R1 to R4	structurally blocked by time; the assumed wall
false humanities (sybil on Embers)	medium	sponsorship capped at two per year, sponsor Kleos at stake, presence proof each era	depends on the size of the established web; the first years are the most exposed
regulatory frame (MiCA, Travel Rule)	high	the Lumen: provable disclosure, native data minimization; payment-currency strategy	structural, not eliminable; readability is the maximum reachable
strict-cap failure (long-term security)	medium	payment volume as the fee resource; public security budget tracking; constitutional tail safety valve	the assumed wager of the symbolic contract
biometric debt	none	no body data collected, anywhere, ever	none; nothing to leak

14.1 Two captured specimens

First specimen, lived on XelisVault on September 5, 2026. The paper wallet generated a twenty-five word mnemonic seed; imported into the official NERVA wallet, it restored an address different from the printed one. The code looked right, re-read twice, and that is precisely the trap: the bug lived in a reading convention, invisible to re-reading. The C++ reference groups the seed into four-byte words and writes them in little-endian order; our TypeScript mirror read them in the opposite order. The fix was made the way it must always be made: port the reference line by line, generate forty-eight test vectors, require bit to bit identity of the two implementations, then re-run the audit. The lesson exceeds the case: an encoding fault is not found, it is crossed; only a second implementation, executed in parallel on the same inputs, reveals it mechanically.

Second specimen, deeper, since it concerns the design itself and not one line of code: the Kleos simulation attack, told in Section 9. A farm of fake profiles crossed the candidacy threshold for the Ring before the honest network and captured the fifty-five seats; four corrective rules closed the window, and the same attack replayed takes no seat. The decisive point is not that the flaw was found: it is that it was found by a three-hundred-line simulation, before any chain code, at a cost in hours of computation and not in years of reputation. The deterministic simulation of a specification is the cheapest machine there is for catching design faults; every new rule of the protocol will be treated that way before implementation, with a fixed seed, numeric invariants and a regression run over three attack sets.

14.2 The six-layer verification protocol

Those two specimens define the method better than a treatise would; it fits in six layers, each extinguishing an entire class of faults rather than one particular bug. A bug of an extinguished class does not come back; a fixed bug always comes back. First layer: cross double implementation on test vectors, which reveals encoding and convention faults. Second layer: deterministic simulation of the social and economic rules before any code, which reveals design flaws. Third layer: reproducible builds and continuous massive

random testing, which reveal non-determinisms and unexpected states. Fourth layer: systematic cross-review between the project holder and his assistant, two pairs of eyes, two reading conventions. Fifth layer: external audit of the differential at every phase, because a fresh auditor does not have the author's blind spots. Sixth layer: measurable GO and NO-GO exit criteria at every roadmap phase, with the rule that a NO-GO remains an acceptable, published result.

The implementation discipline adds to the six layers: never a main network on release-candidate foundations; multi-platform distribution from day one, five signed targets; refusal of unaudited exotic cryptographic primitives, stated since the first version of this specification and held since; publication of all code from the specification phase onward, because hidden code is code nobody has looked at. The method does not promise perfection; it promises the shortest possible reaction time between the appearance of a fault and its capture, and that is the only promise an honest engineer can make.

15 Roadmap and Conclusion

The calendar is realistic because it is sized for a small team, not for a foundation. The base is an assembly of proven foundations, the CryptoNote lineage for the private sphere and the public literature of fast-converging DAGs for the ordering layer; everything that is new, Kleos, the Ring, the Lumen, Cipher perimeters, Mandates, is deterministic, testable, auditable code, with no exotic primitive. Every phase carries one measurable deliverable and one binary exit criterion; an unmet criterion is not a shameful failure, it is steering data, and the phase replays until it is met.

Table 18: Eighteen months, six phases, one binary exit criterion at every step.

Phase	Months	Measurable deliverables	Exit criterion
1, specification	M1-M2	written protocol, numbered architecture decisions, format sketches, extended social-core simulation	specification read by two external readers
2, DAG prototype	M3-M5	dev network mining a CPU DAG at two seconds, stem propagation	twenty-four hours without unexpected reorganization
3, Ring and Kleos v0	M6-M8	signed checkpoints, Deed score computed, era rotation replayed	finality under six seconds over one hundred thousand replayed blocks
4, identities and Lumen	M9-M12	Embers with sponsorship, Ciphers with perimeters, bounded viewing keys, external audit of the differential	audit without a critical flaw
5, public test network	M13-M15	faucet, explorer, one hundred nodes, agent load simulator, upgrade ritual repeated three times	three consecutive upgrades without incident
6, genesis	M16-M18	public ceremony, signed executables for five platforms, Lumen documentation for authorities published	all previous criteria, without exception

Three ecosystem commitments accompany the calendar. The code is public from the specification phase onward, and this whitepaper lives next to it in the same repository, revised like code. The XelisVault ecosystem, register, tickets, paper wallet, becomes multi-chain at phase five: a feature flag, not a rewrite, NERVA first, ANTUMBRA after. Finally, the project identities, domain, repositories, social networks, are reserved before phase two, so that nothing lends itself to confusion on genesis day. The roadmap reads as a promise of operational humility: eighteen months of measured work, every step verifiable by a third party, and none of it depending on a miracle of cryptography.

A faithfully kept shadow, a ring of light anyone can verify on demand: privacy as a right, proof as a courtesy, and time as the only justice of the peace that money cannot corrupt.

Such is the public specification of the ANTUMBRA network. The demand of 2026 is real and measurable: agents that pay, bots to distinguish from persons, authorities to whom one shows proofs rather than promises, users who want the instant. The answer fits one image, the one that gives the network its name: the antumbra, the zone of the annular eclipse where a source too small to be hidden lets a ring of light show around the shadow. The core of the network is that shadow, balances, identities, flows, kept by construction; around it, the ring of verification, reputation, finality, compliance proofs, which anyone can switch on without exposing anyone other than oneself. The specification was written to be attacked before being implemented, and that is the only honest way to write a whitepaper. To work.

References

1. S. Nakamoto. *Bitcoin: A Peer-to-Peer Electronic Cash System*. bitcoin.org, 2008.
2. N. van Saberhagen. *CryptoNote v 2.0*. 2013.
3. tevador et al. *RandomX Proof of Work*. randomx.io, 2019.
4. Y. Sompolinsky, S. Wyborski, Y. Zohar. *PHANTOM and GHOSTDAG: A Scalable Generalization of Nakamoto Consensus*. 2021.
5. B. Bünz, S. Agrawal, S. Boneh, M. Weston. *Bulletproofs: Short Proofs for Confidential Transactions and More*. IEEE S&P, 2018.
6. S. B. Venkatakrisnan, G. Fanti, M. Zhou, K. Cai, P. Viswanath et al. *Dandelion++: Lightweight Cryptocurrency Networking with Formal Anonymity Guarantees*. ACM ToMPECS, 2018.
7. T. Pedersen. *Non-Interactive and Information-Theoretic Secure Verifiable Secret Sharing*. EURO-CRYPT, 1991.
8. M. Castro, B. Liskov. *Practical Byzantine Fault Tolerance*. OSDI, 1999.
9. D. Hopwood et al. *Zcash Protocol Specification*. zips.z.cash, ongoing.
10. Regulation (EU) 2016/679 (General Data Protection Regulation), Article 5(1)(c), data minimization.
11. Regulation (EU) 2023/1114 (Markets in Crypto-Assets, MiCA).